Walmart Sales Analysis

SQL-Enhanced Business Analysis Report

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Purpose: Identify underperforming stores, understand weekly sales patterns, compare holiday and non-holiday performance, and examine whether external factors are associated with sales.

SQL dialect: Google BigQuery Standard SQL. The queries use a table named `walmart`; replace this with your actual project.dataset.table name if your data is stored elsewhere.

Important: The numeric findings below are carried over from the supplied report. The SQL is provided to make each finding reproducible; run it against your loaded dataset to confirm the values before publishing.

1. Dataset Overview

Attribute	Reported details
Rows	6,435
Columns	8
Stores	45
Date range	5 February 2010 – 26 October 2012
Observation level	Weekly sales for each store

2. Data Dictionary

Column	Meaning	Type
Store	Store identification number	Integer
Date	Weekly sales reporting date	Date
Weekly_Sales	Total sales for a store during that week	Numeric
Holiday_Flag	1 = holiday week; 0 = non-holiday week	Binary
Temperature	Temperature during the week	Numeric
Fuel_Price	Fuel price during the week	Numeric
CPI	Consumer Price Index	Numeric
Unemployment	Unemployment rate	Numeric

3. Data Quality Assessment

The original report records zero missing values, zero duplicate rows, and zero invalid values, with the date field converted to a date type and records sorted by store and date. Use these queries to check the data after loading it.

3.1 Row count, date range, and store count

```
SELECT
  COUNT(*) AS total_rows,
  COUNT(DISTINCT Store) AS store_count,
  MIN(Date) AS first_date,
  MAX(Date) AS last_date
```

```
FROM `walmart`;
```

3.2 Missing values

```
SELECT
  COUNTIF(Store IS NULL) AS missing_store,
  COUNTIF(Date IS NULL) AS missing_date,
  COUNTIF(Weekly_Sales IS NULL) AS missing_weekly_sales,
  COUNTIF(Holiday_Flag IS NULL) AS missing_holiday_flag,
  COUNTIF(Temperature IS NULL) AS missing_temperature,
  COUNTIF(Fuel_Price IS NULL) AS missing_fuel_price,
  COUNTIF(CPI IS NULL) AS missing_cpi,
  COUNTIF(Unemployment IS NULL) AS missing_unemployment
FROM `walmart`;
```

3.3 Duplicate rows

```
SELECT
  COUNT(*) - COUNT(DISTINCT TO_JSON_STRING(t)) AS duplicate_rows
FROM `walmart` AS t;
```

This duplicate check treats fully identical rows as duplicates. If your table contains additional columns, the query checks those too.

3.4 Invalid values / date checks

```
SELECT
  COUNTIF(Store <= 0) AS invalid_store_ids,
  COUNTIF(Weekly_Sales < 0) AS negative_sales_rows,
  COUNTIF(Holiday_Flag NOT IN (0, 1)) AS invalid_holiday_flags,
  COUNTIF(Date IS NULL) AS invalid_dates
FROM `walmart`;
```

Sorting is a query-level operation; use ORDER BY when displaying records. BigQuery tables do not retain a guaranteed row order.

```
SELECT *
FROM `walmart`
ORDER BY Store, Date;
```

4. Initial Business Metrics

4.1 Overall sales, average store-week sales, and holiday records

```
SELECT
  SUM(Weekly_Sales) AS total_sales,
  AVG(Weekly_Sales) AS avg_store_week_sales,
  COUNTIF(Holiday_Flag = 1) AS holiday_records,
  COUNTIF(Holiday_Flag = 0) AS non_holiday_records
FROM `walmart`;
```

Reported outcomes: total sales ≈ $6.737 billion; average store-week sales ≈ $1.047 million; holiday records = 450; non-holiday records = 5,985.

5. Business Questions, Outcomes, and SQL

5.1 Which stores generate the highest and lowest sales?

Reported outcome: Store 20 has the highest total sales (about $301.40 million); Store 33 has the lowest (about $37.16 million).

```
SELECT
  Store,
  SUM(Weekly_Sales) AS total_sales
FROM `walmart`
GROUP BY Store
ORDER BY total_sales DESC;
```

The first row is the highest-sales store; the last row is the lowest-sales store. For a compact output, add LIMIT 1 to a descending query for the highest store and use ASC for the lowest.

5.2 How do sales change over time?

Reported outcome: highest-sales week was 24 December 2010 (about $80.93 million); lowest-sales week was 28 January 2011 (about $39.60 million). The overall trend was relatively stable with noticeable peaks.

```
SELECT
  Date,
  SUM(Weekly_Sales) AS total_weekly_sales
FROM `walmart`
GROUP BY Date
```

```
ORDER BY total_weekly_sales DESC;
```

For the lowest-sales week, change the final line to ORDER BY total_weekly_sales ASC. This sums sales across all stores for each reporting date.

```
SELECT
  Date,
  SUM(Weekly_Sales) AS total_weekly_sales
FROM `walmart`
GROUP BY Date
ORDER BY Date;
```

5.3 Do holiday weeks generate higher sales than non-holiday weeks?

Reported outcome: holiday dates: 10; non-holiday dates: 133. Average total sales across stores per holiday date ≈ $50.53 million, compared with ≈ $46.86 million for non-holiday dates.

```
WITH date_level AS (
  SELECT
    Date,
    MAX(Holiday_Flag) AS holiday_flag,
    SUM(Weekly_Sales) AS total_sales_for_date
  FROM `walmart`
  GROUP BY Date
)
SELECT
  IF(holiday_flag = 1, 'Holiday', 'Non-Holiday') AS week_type,
  COUNT(*) AS number_of_dates,
  SUM(total_sales_for_date) AS total_sales,
  AVG(total_sales_for_date) AS average_sales_per_date
FROM date_level
GROUP BY week_type
ORDER BY week_type;
```

This query aggregates across stores first, so it compares total Walmart sales for each date rather than treating every store row as a separate week.

5.4 Which stores have the most consistent sales?

Reported outcome: Store 3 was described as the most consistent, with average sales around $518,900. Consistency should be measured by variation (for example, standard deviation or coefficient of variation), not by average sales alone; the query below ranks stores by coefficient of variation.

```
SELECT
  Store,
  AVG(Weekly_Sales) AS average_weekly_sales,
  STDDEV_POP(Weekly_Sales) AS sales_stddev,
  SAFE_DIVIDE(STDDEV_POP(Weekly_Sales), AVG(Weekly_Sales))
    AS coefficient_of_variation
FROM `walmart`
GROUP BY Store
ORDER BY coefficient_of_variation ASC;
```

The first row is the least variable relative to its average. Run this query to verify whether Store 3 is indeed the most consistent under this definition.

5.5 Is there a relationship between temperature and sales?

Reported outcome: temperature has a very weak negative correlation with weekly sales.

```
SELECT
  CORR(Temperature, Weekly_Sales) AS temperature_sales_correlation
FROM `walmart`;
```

5.6 Does fuel price have any relationship with sales?

Reported outcome: fuel price shows almost no linear relationship with weekly sales.

```
SELECT
  CORR(Fuel_Price, Weekly_Sales) AS fuel_price_sales_correlation
FROM `walmart`;
```

5.7 How do CPI and unemployment relate to sales?

Reported outcome: CPI does not appear to have a meaningful direct relationship with sales. As unemployment increases, sales tend to decrease slightly, but the relationship is not strong.

```
SELECT
  CORR(CPI, Weekly_Sales) AS cpi_sales_correlation,
  CORR(Unemployment, Weekly_Sales) AS unemployment_sales_correlation
```

```
FROM `walmart`;
```

Correlation indicates linear association, not causation. Store-level differences, seasonality, promotions, inventory, and customer mix may also affect sales.

6. Key Findings

• **Store performance:** Store 20 is reported as the highest-sales store and Store 33 as the lowest-sales store.

• **Time pattern:** sales are generally stable, with peaks around particular dates.

• **Holiday effect:** average total sales per holiday date are reported to be higher than per non-holiday date.

• **External factors:** temperature and fuel price show weak relationships with weekly sales; CPI and unemployment relationships are also reported as weak.

• **Consistency:** use standard deviation or coefficient of variation to identify stable stores; verify the original Store 3 claim with the supplied query.

7. Recommendations

• **Investigate Store 33:** review product mix, customer demand, inventory availability, location, promotions, and operating performance.

• **Benchmark against Store 20:** compare sales patterns, assortment, execution, and local conditions.

• **Plan for seasonal peaks:** align inventory, promotions, and staffing with historically strong periods.

• **Monitor weekly performance:** maintain a dashboard showing total sales, store rankings, trends, and holiday comparisons.

• **Look beyond external variables:** because the measured relationships are weak, investigate store-specific operational and customer-related factors.

8. SQL Summary / Query Index

Report outcome	Query / SQL function
Dataset size and date range	COUNT(*), COUNT(DISTINCT Store), MIN(Date), MAX(Date)
Missing values	COUNTIF(column IS NULL)
Duplicate rows	COUNT(*) and COUNT(DISTINCT TO_JSON_STRING(row))
Total and average sales	SUM(Weekly_Sales), AVG(Weekly_Sales)
Top / bottom stores	GROUP BY Store; ORDER BY SUM(Weekly_Sales)
Highest / lowest sales date	GROUP BY Date; ORDER BY SUM(Weekly_Sales)
Holiday comparison	Date-level aggregation; AVG(total sales per date)
Sales consistency	STDDEV_POP and coefficient of variation
External-factor relationships	CORR(variable, Weekly_Sales)